# The Youth Science Journal Of Iraq

Issue #4 January 2026

# NEW YEAR SPECIAL



Satvik M. Bhure

# The Youth Science Journal Of Iraq

## YSJOI ISSUE #4

**©2026 Satvik M. Bhure**

**First Edition – January 31st 2026**

**No part of this publication may be copied or reproduced without written permission from the author.**

**Contact for permissions at YSJOIofficial@gmail.com**

**Table of Contents:**

**Cover**

**Copyright**

**A Short Editorial**

**Question I: If time can slow down or speed up, can it ever stop completely?**

**Question II: Why does the universe seem to follow math so perfectly?**

**Question III: Could there be life that doesn't use water at all?**

**Question IV: If everything is made of particles, what holds reality together at the smallest scale?**

**Question V: Why is the speed of light the speed limit of the universe?**

**A New Year's Gift**

**Acknowledgements**

**Conclusion**

## A Short Editorial:

### YSJOI - by the youth, for the youth of Iraq!

In the great land that was once Mesopotamia, and the home of the hanging gardens of Babylon, progress in science has slowed.

Why?

Because the youth of Iraq have no platform to build upon! Opportunities to learn about science, technology, engineering, and mathematics are rare, and students trying to learn about them are even actively discouraged – I speak from experience.

But instead of surrendering to that silence, I have chosen to rally the youth of Iraq. If we do not have a base to stand on, we will build one!

And the first step to that is this Youth Science Journal Of Iraq, YSJOI, where we aim to provide that very base – an opportunity to teach, learn and explore through different articles – be it inspiring them, writing them, or reading them!

Thank you for being part of this beginning.
Now, dive in – into the weird, the beautiful, and the wonderful.

– Satvik M. Bhure, Founder of YSJOI

## Question I: If time can slow down or speed up, can it ever stop completely?

*– Alina Parwar, 14*

Time! The ethereal thing that we have only just begun to quantify, and might never be able to fully categorize. Is it a quantity? Is it a dimension? And whatever it is, how does it change speed around certain objects?

Most of us know about time dilation by now, the way that time seems to stretch out or slow down around some things – like black holes, for example. We know it can stretch, slow, and warp – Einstein proved that much. A fast-moving astronaut ages more slowly than someone on Earth. A clock near a massive object ticks more sluggishly than one far away. So it's natural to ask the dangerous follow-up: if time can slow down… could it ever hit zero?

From the outside, it almost looks like it can. Near a black hole, gravity becomes so intense that time slows dramatically. To a distant observer watching from afar, an object falling toward the black hole's edge – the event horizon – appears to slow down more and more, never quite crossing it. Its clock ticks slower, its light stretches redder, until it seems frozen in time. From *our* perspective, time never fully stops for that object – but it asymptotically crawls toward a standstill, always almost at zero, but never quite hitting it, until the light of the object is redshifted too far and the object seems to just… disappear. But here's the twist: for the falling object itself, time doesn't stop at all. Its own clock ticks normally. It crosses the event horizon without noticing anything special. Well, except being destroyed, of course; but nonetheless this tells us something profound: time stopping isn't an absolute event – it depends on who's watching. There is no universal pause button for the cosmos, only perspectives stitched together by spacetime.

Could time stop *everywhere*, for everyone? Maybe not – but here's the catch. If time stopped, the universe would stop also; so scarily enough, we wouldn't notice at all. For time to stop, all light and particles must stop with it, meaning that even if it did happen… we would never know.

## Question II: Why does the universe seem to follow math so perfectly?

*– Melina Khalid, 14*

Quite literally taking on the entire subject matter of STEM – science, technology, engineering, and math – in one question? Incredible. So why does the universe seem to fit our mathematical ideals so well? Why do flowers follow the Golden Ratio, and why does the light of the sun *never* exceed a speed that we could not predict?

Well, the answer is partly breaking the question itself – but as it is, the universe does not follow math! Math follows the universe, because every branch of human study – be it the all-important sciences or the ever-interesting philosophy – is an extension of humanity's need to understand the universe! We did not invent math, and find that the universe seems to follow our equations for kinetic energy and our predictions for the speed of light – instead, we calculated, over centuries, and discovered a way to interpret the universe, and named it Mathematics; because be it differentiation of a student's tenth year or the confusing relations of quarks at a quantum level; be it Physics or Chemistry or the study of the nucleus in biology – it all relies on our understanding of math, because through it we can describe the very universe itself. Many today question, indeed, how the basic things we learn in elementary today could possibly be at all important to the universe, grand and mystic that it is; we often forget that even the simple counting we learn as babies comes from a Base-10 system that once had no concept of zero; the works of Indian, Mesopotamian, and Greek mathematicians all combined over centuries before we could perform even the most basic calculations today. What we think of as the equation which is *so* famous today as the cumulative work of Einstein – the basis of special relativity – $E=mc^2$ – is a simplified explanation of a much larger explanation of the same equation - $E^2=(m_0c^2)^2+(pc)^2$, which itself came from years of calculation.

So next time we wonder how math describes the world so well and how the world lets itself be described so well, let us remember the millennia of work that was put in before we could learn even to count today!

## Question III: Could there be life that doesn't use water at all?

*– Rami Karwan, 16*

Water is life – or at least, life as we know it. Every plant, animal, microbe, and human is drenched in it. Our cells, our blood, our oceans – all proof that life loves $H_2O$. But could life exist that doesn't? Could organisms arise that never touch a single drop of water?

The short answer: maybe. But it would have to be extremely different from us. Water is special because it dissolves things, allows chemical reactions, and transports molecules – basically, it's nature's almost universal solvent. Life needs chemistry, reactions, energy transfer; and water makes it all possible. Without it, some other liquid would have to step in. Scientists imagine alternatives; liquid ammonia, methane, or even supercritical carbon dioxide, as possible "solvents" for alien life. On Titan, Saturn's moon, rivers of methane carve icy valleys – maybe tiny microbes, entirely unlike anything on Earth, could exist there. After all, there isn't a rulebook out in nature that says that Earth's life is perfect, or that all other life that exists should copy us.

However, there is, in fact a small issue; these alternative chemistries are slower, less versatile, and probably wouldn't allow the fast, efficient life we're used to, and evolution tends to go with the best possible way that it can find. Of course, in a pressured environment, where water is not so easily found, evolution, ever the tinkerer, might figure it out in a billion years. Life elsewhere might breathe liquid methane, metabolize nitrogen instead of carbon, or swim in oceans we'd call poisonous.

In the end, water isn't a universal requirement – it's just what life on Earth discovered first. The cosmos might hide creatures so strange that we wouldn't even recognize them as "alive." And that's the beauty of imagination fused with science: anywhere there's chemistry, there might just be life, in forms we can't yet dream of, more beautiful and scarier than anything we even know of.

## Question IV: If everything is made of particles, what holds reality together at the smallest scale?

*– Muhammed Hemin, 15*

Everything! Your pencil, the chair you're sitting on, the air around you, even you yourself; made of tiny particles, each smaller than the smallest dust mote, each buzzing with energy. But here's the kicker: particles are mostly empty space. If atoms are tiny solar systems, the nucleus is the sun and electrons are planets – but mostly, it's space. So why doesn't everything collapse? Why isn't reality just a cloud of ghosts?

The answer lies in forces – four fundamental forces that stitch the universe together. Electromagnetism keeps electrons orbiting nuclei and allows chemistry to happen. The strong nuclear force glues protons and neutrons into atomic cores. The weak nuclear force powers the stars and radioactive decay. And gravity? Well, gravity binds planets, stars, galaxies – and even though it's weak at atomic scales, it shapes the cosmos.

At the tiniest scale, reality is also held together by quantum fields. Particles aren't solid balls – they're excitations, ripples in invisible fields. The electron is a wave in the electron field; photons are ripples in the electromagnetic field. Forces are just messengers, carrying energy and momentum between particles. These fields are the threads of the universe's tapestry. Without them, there's no stickiness, no chemistry, no "everything" at all for the question to ask about.

So at the bottom, reality isn't made of things – it's made of relationships. And as cheesy as that sounds, particles, fields, and forces dance together in an eternal choreography, forming everything from dust to stars to you. It's a universe held together by invisible threads – like magic, only mathematical and real.

Could you say that gluons hold together quarks and neutrons keeps protons from flying apart? Maybe that would be technically right, but what about at larger scales and the medium scales? Everywhere we look, particles, objects and entire galaxies are held together by their own essence – by a magic older than any human magic.

# Why is the speed of light the speed limit of the universe?

– *Ronya Huner, 12*

Nothing can go faster than light. Nothing. Not spaceships, not information, not even your thoughts, and unfortunately, not our fingers when our homework is due and we haven't yet completed it. But why?

The speed of light isn't arbitrary or accidental. If we look at special relativity and look at the way the universe behaves, the speed of light is baked into the fabric of spacetime itself. Einstein taught us that space and time are intertwined; moving through one affects the other. As objects approach light speed, their mass appears to increase, and time slows down for them. To reach light speed, a massive object would need infinite energy – which is prohibited by the laws of physics, and so is literally impossible. Light, being massless, can travel at this speed freely, and that speed becomes the ultimate cosmic constant, the benchmark for many, many of our achievements – maybe, just maybe, one day even leading us to time travel.

Why does the universe impose a speed limit at all? Would everything not be so much more efficient if light was instant, flying from place to place like it is so often portrayed in sci-fi? Well, to answer that, we must first think of spacetime like a trampoline. The speed of light is how fast ripples can move across it. Push faster, and the trampoline tears; reality itself doesn't allow it. This limit preserves cause and effect – without it, you could send information backward in time, and paradoxes would explode everywhere. The speed of light is less a limit than a rule of the cosmic game. Just as we cannot travel through the fourth dimension as third dimensional beings, light cannot travel faster than this limit in our three dimensional world.

And here's a fun twist: even though nothing surpasses light, the universe itself expands faster than light in some places. Galaxies drift apart at speeds beyond our imagination, yet no laws are broken – space itself is stretching, carrying them along. The cosmic speed limit applies to objects *within* space, not to space itself.

So is it still a speed limit, or just a challenge for our brightest minds?

**A New Year's Gift**:

With the advent of the new year, today, with the first YSJOI issue of 2026, we wish to give you all a small surprise! A calendar of all the celestial events visible from Iraq this year, listed for your pleasure (which you can print out directly from here!)

# 🗓️ 2026 Skywatching Calendar – Visible from Iraq (Chronological)

## January

1. **Jan 3–4 – Quadrantid Meteor Shower (Peak)**
   One of the first major meteor showers of the year; best before dawn.

2. **Jan 10 – Jupiter at Opposition**
   Jupiter shines brightest of the year, visible all night.

---

## March

3. **Mar 3 – Total Lunar Eclipse**
   The Moon enters Earth's shadow and can turn reddish – visible at night from Iraq.

---

## April

4. **Apr 22–23 – Lyrid Meteor Shower (Peak)**
A spring meteor shower visible late night into early morning.

---

## May

5. **May 5–6 – Eta Aquariid Meteor Shower (Peak)**
Debris from Halley's Comet streaks across the sky before dawn.

---

## August

6. **Aug 12–13 – Perseid Meteor Shower (Peak)**
One of the best meteor showers of the year, excellent viewing after midnight.

7. **Aug 28 – Partial Lunar Eclipse**
A deep partial eclipse of the Moon, visible where the Moon is above the horizon (Iraq included).

---

## 🔭 Other Notable 2026 Sky Events *(also visible from Iraq, though not dramatic "events" like eclipses)*

- **June 6–7 – Venus–Jupiter Conjunction** – Two bright planets appear close together after sunset.
- **Oct 21–22 – Orionid Meteor Shower (Peak)** – Halley's Comet debris produces meteors late night.
- **Nov 17–18 – Leonid Meteor Shower (Peak)** – Moderate storm, early morning best.
- **Dec 13–14 – Geminid Meteor Shower (Peak)** – One of the most prolific meteor showers of the year.

Acknowledgements

We thank all the scientists who have worked thus far to find all about the world so we can learn about it, and everyone who has supported any member of the YSJOI team in furthering our mission.

Special thanks to:

Ronya Huner

Muhammed Hemin

Alina Parwar

Melina Khalid

Rami Karwan

For these incredible questions and giving us a chance to answer them in the January 2025 issue of YSJOI!

And we thank you, reader, for supporting our mission to help the Iraqi youth where there has been no help as yet!

# CONCLUSION

We hope to further YSJOI's mission to empower the youth of Iraq in STEM fields by our monthly magazine, and sincerely hope we were able to help you and other students with this monthly issue! We are always open to feedback and questions, and everyone is of course welcome to learn in their own way; after all, we are all different, and we all belong.

Make sure to visit our website at https://YSJOI.is-great.org

You can email us at YSJOIofficial@gmail.com

Or DM us at www.instagram.com/ysjoi_official